

Capstone Milestone 2: Data Summary

Willamette Applicant Trajectory: Predicting Student College Enrollment Destination

Seira Ramchandani

Courtney St. Onge

2026-06-29

Table of contents

1	1. Resarch Question Reminder	1
2	2. Data Inventory	2
3	3. Schema and Organization	5
3.1	ERD	5
4	4. Ingestion Status	6
4.0.1	Missingness	7
4.0.2	Additional Cleaning Steps	8
5	6. Reproducibility	9
5.0.1	Prerequisites	9
5.0.2	Execution Order	9
6	7: Ethics and Access Controls	10
7	8. Freeze Statement	10
7.0.1	Deltas from M1 proposal	10

1 1. Resarch Question Reminder

This project investigates what applicant characteristics predict whether a student admitted but not enrolled at Willamette University will enroll at a fellow private NW5 institution (Reed College, Lewis & Clark College, Whitman College, or University of Puget Sound), a large public university, or another destination category that may emerge from the data. The analysis

draws on a frozen dataset spanning three complete application cycles (2023-2025) sourced from *Willamette's Technolutions Slate CRM* and matched to *National Student Clearinghouse enrollment records*. This scope was established in the project proposal and remains unchanged, providing a sufficient historical window to train and evaluate classification and clustering models and will be used by the Admissions Office to segment incoming applicants and enable targeted outreach strategies accordingly.

2 2. Data Inventory

Dataset	Source	Row Counts	Date Range	Refresh Policy	Repo Location
Slate applicant data	Technolutions Slate CRM (SFTP)	9,404	2023-2025	Historical pull covering a 3-year window. Frozen.	data/raw/willamette_app_detail
NSC enrollment 2023	National Student Clearinghouse (CSV)	2,353	2023	Historical pull. Frozen.	data/raw/nsc_fall23.xlsx
NSC enrollment 2024	National Student Clearinghouse (CSV)	2,120	2024	Historical pull. Frozen.	data/raw/nsc_fall24.xlsx
NSC enrollment 2025	National Student Clearinghouse (CSV)	2,125	2025	Historical pull. Frozen.	data/raw/nsc_fall25.xlsx
Carnegie Classification of Institutions of Higher Education	Website URL **	3928	2025	Manual pull. Static reference data. Frozen.	data/external/carnegie_classification

** External CSV file downloaded from Carnegie resource website and columns dropped to consist of only unitid, instnm, stabbr, ic2025size. This dataset will be used to categorize the

destination institutions of non-Willamette enrollees as large public or other. This will be a step in the feature engineering process, so is not included in the ingestion or cleaning scripts.

3 3. Schema and Organization

3.1 ERD

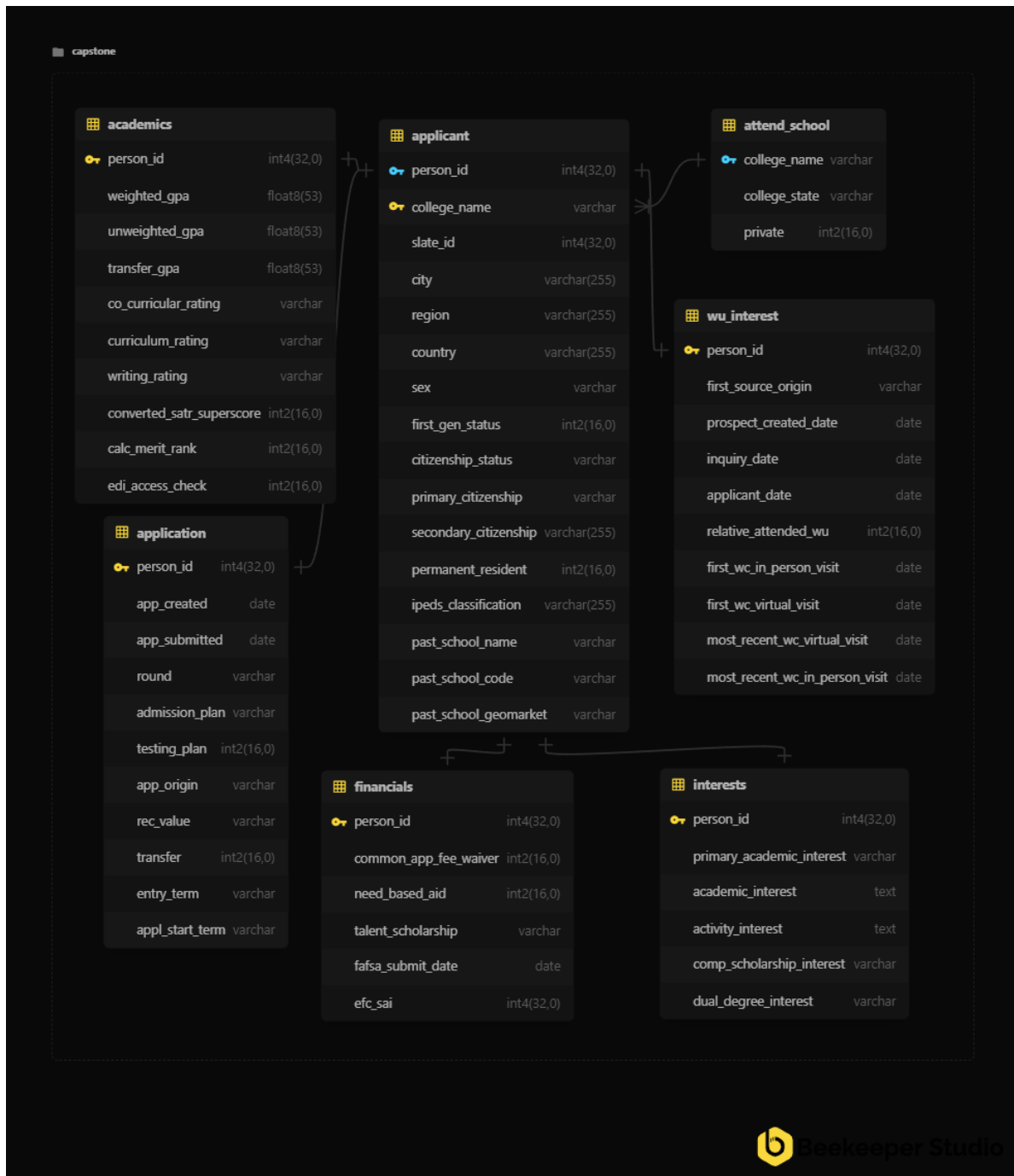


Figure 1: ERD

The normalized ERD above illustrates the logical structure of the Slate CRM staging tables and NSC staging table, which serves as a reference for understanding the data's organization. All seven tables share `person_id` as the primary key, with `applicant` and `attend_school` additionally joined on `college_name`. These staging tables inform the structure of the analysis-ready file but are not themselves used directly in modeling.

The analysis-ready dataset is a flat CSV currently residing in `data/interim/` while cleaning is ongoing, and will be moved to `data/processed/` upon completion. It was constructed by an inner join of the Slate applicant data onto the NSC enrollment data on `a.colleague_id = n.appl_students_id`, a 7-digit unique student identifier named differently across the two systems but representing the same person. This join intentionally scopes the dataset to non-Willamette enrollees with confirmed destination data, dropping both students who opted out of NSC tracking and students who enrolled at Willamette, as the research question focuses on predicting where non-enrollees go. Before joining the two data sources, duplicate student records in the NSC data were removed by retaining the earliest `appl_start_term` per student, treating a student's first recorded enrollment destination as ground truth. The resulting flat file contains **6,558** rows.

The data layers relate as follows: - `data/raw/` : original CSV exports as received from the SFTP server, never modified - `data/external/` : static reference data, never modified - `data/interim/` : current working stage; joined and deduplicated flat file in progress - `data/processed/` : final analysis-ready dataset, to be committed upon completion of cleaning

4 4. Ingestion Status

Data ingestion was completed as a one-time manual pull from the SFTP server, consistent with the frozen data scope of this project. The Slate CRM export was generated via a query co-developed with the Admissions stakeholder and deposited into a shared SFTP server. The NSC enrollment data was similarly delivered as three separate CSV files and retrieved from the same server. The raw CSV files in `data/raw/` represent the complete and unmodified output of the ingestion query.

Access to the SFTP server requires three security controls: a raw password, a public/private key pair held exclusively on Seira and Courtney's local devices, and an IP address whitelist restricted to the Willamette University campus network. Data was transferred from the server to local machines using FileZilla. As a result of these access constraints, re-ingestion requires both a team member's credentialed local device and a connection to the Willamette campus network. This is a documented constraint rather than a gap, and is appropriate given the sensitivity of the data.

No ingestion failures were encountered. Setup requirements were limited to FileZilla installation on both teammates' local devices and confirmation of IP whitelisting. Since the data is frozen, no scheduled orchestration or automated pipeline is in place or needed.

#5. Data Quality and Profiling

Check	Result
Duplicate records	0
Missing predictor values	See table below
Unmatched rows after join	40 rows. 0.6% (details listed in section 3)
Date range validated	Fall2023 - Fall2025 entry cohorts

4.0.1 Missingness

Column	Missing Count	Missing %	Notes
app_transfer_gpa	6,444	98.3%	Expected; most applicants are first-year, not transfer students
relative_attended_w6	6,280	95.8%	Nulls will be recoded to 0.
permanent_resident	6,229	95.0%	Expected; most applicants are US citizens
converted_satr_superscore	3,941	60.1%	Expected; reflects test-optional policy

Column	Missing Count	Missing %	Notes
efc_sai	1,860	28.4%	Likely reflects students who did not submit FAFSA; imputation strategy TBD. A missingness indicator column may be added as a model feature, as non-submission of FAFSA may itself be predictive
rec_value	891	13.6%	Likely reflects incomplete applications
common_app_fee_value	219	3.34%	Minor; likely incomplete applications
school_1_code	87	1.33%	Minor; likely international students with non-standard records
app_weighted_gpa	61	0.93%	Minor
app_unweighted_gpa	60	0.91%	Minor
need_based_aid	43	0.66%	Minor
first_gen_status	37	0.56%	Minor
testing_plan	1	0.02%	
All other columns	0	0%	

4.0.2 Additional Cleaning Steps

Several columns required encoding and standardization during cleaning. This includes conversion of binary text fields to smallint (0/1), conversion of ordinal text ratings to numeric scales, and high school name standardization across ~80+ records using school code as the authoritative identifier Full detail on column-level transformations is documented in the data cleaning

notebook (notebooks/data_cleaning.md) and the data dictionary (data/SCHEMA.md).

5 6. Reproducibility

The analysis-ready dataset can be rebuilt from the raw files using the following steps:

5.0.1 Prerequisites

- PostgreSQL database accessible via Beekeeper Studio
- Raw CSV files transferred from the SFTP server into a local directory, as described in Section 4
- The staging schema must be created and raw files imported as tables in Beekeeper using New Table from File before any scripts are run. Values must be nullable, and there are a few datatypes that should be adjusted from the automatic detection. The following data types for import should be:
 - academic_interest: text
 - activity_interest: text
 - school_1_code: varchar(255)
 - person_efc/sai: float

```
CREATE SCHEMA staging;  
CREATE TABLE staging.wu_app (  
    LIKE public.wu_app_details INCLUDING ALL  
);
```

5.0.2 Execution Order

All scripts are located in src/Capstone_data_cleaning/ and should be run in the following order in Beekeeper Studio: 1. schema_creation.sql : creates the capstone schema and all normalized tables 2. nsc_cleaning.sql : cleans and standardizes the NSC staging tables and unions them across all three application cycles 3. wu_app_cleaning.sql : cleans and standardizes the Slate applicant staging table 4. data_migration.sql : migrates cleaned data from the staging schema into the capstone schema 5. all_data_flat_file_creation.sql : joins all data into a single flat file for analysis and use in machine learning techniques

The order of steps 2 and 3 is interchangeable. No specific PostgreSQL version is required, as scripts have been tested across team members' environments. Once the script has been run, the resulting table is exported as a CSV from Beekeeper Studio and moved manually into data/interim/.

6 7: Ethics and Access Controls

All PII was removed prior to data landing in data/raw/, ensuring that no personally identifiable information is present at any stage of the repository. In the original raw data, birthdate, sex and gender were present, which we took note of as a combination that could reasonably enable re-identification of a student even in the absence of direct identifiers. The query co-developed within Slate was updated to remove these columns and re-exported into the SFTP server.

Both project owners have signed a formal confidentiality agreement governing the use of this data. Access to the source data is restricted to credentialed team members on the Willamette campus network, as described in Section 4. The data will not be shared with other members of the capstone cohort, external stakeholders beyond the primary stakeholder, or the public. No specific use constraints were imposed by either Technolutions or the National Student Clearinghouse beyond the confidentiality agreement. The data is used solely for the purposes described in the project proposal and will not be repurposed outside of this project scope.

7 8. Freeze Statement

The dataset for this project is now frozen. No further ingestion, additional data pulls, or new data sources are planned. The analysis will proceed exclusively on the datasets described in this summary.

7.0.1 Deltas from M1 proposal

Delta 1: The project scope has been explicitly clarified to focus only on students who were admitted to Willamette University but enrolled elsewhere. Although this was implied in the original project proposal, it was not stated explicitly. This clarification does not change the data used or the research questions.

Delta 2: A new external dataset, the Carnegie Classification of Institutions of Higher Education, was identified as necessary after the initial data freeze to construct the target variables is_nw5 and is_large_public. The size classification (ic2025size) categorizes institutions on a 1–5 scale based on total 12-month headcount, with categories 4 (Large: 20,000–40,000) and 5 (Very Large: 40,000+) used to identify large institutions, filtered further by public/private status. A join on the NSC data on College Name will occur during the feature engineering stage as the dataset moves from data/interim/ to data/processed/, and is therefore not reflected in the existing ingestion or cleaning scripts.